

Effect of Current on Yield in a Potassium Chlorate Electrolysis Cell

jtb200

Contents

1	Introduction	1
2	Background Research	1
2.1	Principles of a Chlorate Cell	1
2.2	Useful Information about Electric Circuits	2
2.3	Types of Efficiency Measurements	2
3	Procedure	2
3.1	Variables	2
3.2	Safety, Ethical and Environmental Considerations	4
3.3	Preparing the Electrolyte	4
3.4	Selecting the Anode	4
3.5	Construction of the Cell	5
3.6	Trials	5
4	Qualitative Observations and Data	6
5	Data Processing	7
5.1	Current Efficiency	7
5.2	Energy Efficiency	7
5.3	Processed Data	8
6	Conclusion	10
6.1	Possible Sources of Error	11
6.2	Summary	12
7	Works Cited	
8	Appendix	

1 Introduction

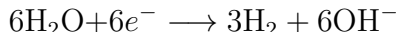
The aim of this investigation is to answer the question "What is the optimal current to produce potassium chlorate with maximum efficiency in an electrolysis cell?". Industrially, electrolysis is the main means of production for potassium chlorate. As electricity is the most valuable resource used to produce it, optimizing a cell's efficiency is greatly beneficial (EPA, 2019). Potassium chlorate is a strong and exciting oxidizer, which finds widespread use in pyrotechnics, bleach production for the manufacture of paper and as a laboratory reagent (American Chemical Society, 2019). The topic is one of great interest to me and I have made potassium chlorate via electrolysis in the past for my own experiments.

2 Background Research

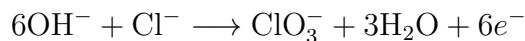
2.1 Principles of a Chlorate Cell

A chlorate cell is an electrolytic cell that converts an electrolyte of chloride salts into chlorate salts. The mechanism through which chlorate is formed is described by the equations below (NurdRage, 2019):

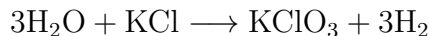
Cathode Reaction:



Anode Reaction:



Overall:



These are only the ideal reactions however; in actuality many more are carried out. The actual reactions depend on many factors such as cell temperature, pH and electric potential. For example, electricity in the cell can also be used to break water into hydrogen and oxygen gas or turn sodium chloride into sodium hydroxide and chlorine gas. There are also several routes through which chlorate can be formed, including first the formation of hypochlorite and then chlorate. The chlorate can even be reduced back to chloride at the cathode.

For this investigation potassium chloride will be used as the chloride salt. The benefit of this approach is that potassium chlorate, the desired product, is relatively insoluble in water,

so it will precipitate out as it forms, allowing accurate yield readings to be taken.

2.2 Useful Information about Electric Circuits

Charge, which is measured in Coulombs, is directly proportional to the amount of electrons that pass through the cell. Charge and current are related by the below equation, where Q is charge, I is current, and t is time (Stanford, n.d.):

$$Q = It$$

Because voltage (or electrical potential) is defined as energy per unit charge, energy used in an electrical circuit can be calculated using the following equation, where V is voltage and E is energy (Harvard, 2015):

$$E = QV$$

In an electrical circuit an increase in current requires an increase in voltage. This is described in Ohm's law, which states that voltage equals current times resistance. While the electrolyte of an electrolysis cell does not follow this law, an increase in current will still require an increase in voltage and visa versa.

2.3 Types of Efficiency Measurements

There are two types of efficiency that will be measured in this experiment: current efficiency and energy efficiency. Current efficiency measures the fraction of electrons that enter the cell used to carry out the desired reaction. Energy efficiency measures the fraction of energy entering the cell that was used to provide the energy change associated with the desired reaction. Both charge efficiency and current efficiency are important in different ways and will both be discussed in this paper.

3 Procedure

3.1 Variables

The independent variable for this investigation was the electrical current through the cell. It was changed in 0.5 ampere intervals, ranging from 0.5A to 3.5A. This was done using a

variable current power supply.

It was decided to control current rather than voltage for two main reasons: current is the same at any point in a circuit and the calculation of charge is easier when current is constant. Since charge passing through the cell was controlled, and charge is current times time, a current controlled cell makes charge control much easier. If the voltage was controlled instead, current would change during the trial and it would thus need to be constantly monitored for an accurate charge calculation to be made.

Because current is the same at any point in a (series) circuit, the current measured by the power supply is the same as the current through the cell. The same does not apply to voltage however, as there was a voltage drop of about 100mv between the cell and the power supply. This means that in order to control voltage across the cell, it would have to be measured at the cell instead of voltage at the power supply. This was considered overly complicated, providing further reason for current control.

The dependent variable for this experiment was the yield of potassium chlorate, measured in grams using an electrical balance. This was then used to calculate both current and energy efficiency.

Some control variables were the concentration of ions in the electrolyte, the volume of electrolyte used, the electrolysis cell used, the temperature of the surroundings, the drying time and temperature for the potassium chlorate, and the total charge passed through the cell.

The choice to control charge proved to have a trade-off as this meant that the run times would vary between trials. As mentioned previously, current is charge per unit time. This means that at higher currents less time is needed to move the same amount of charge. It was decided that charge is a more important variable to control than run time because when charge is controlled, the maximum possible yield of potassium chlorate is always the same. This means that the theoretical chloride concentration at the end of each trial is the same, so any changes in chloride concentration between trials can be attributed to changes in their efficiencies. As ion concentration may have an effect on current efficiency, ensuring that ion concentrations remain mostly constant is important.

A conscious decision was made to not control cell temperature in this investigation. Because higher current cells produced more heat, temperature of the cell would vary between trials. It was decided that since this varying of temperature is intrinsic to the varying of current, temperature should not be controlled as to make this experiment more accurately reflect how the cell would naturally run.

3.2 Safety, Ethical and Environmental Considerations

While they are produced in small quantities, several harmful substances are inevitably encountered during the operation of a chlorate cell. These harmful substances are chlorine gas, potassium hypochlorite, and potassium hydroxide. To mitigate risks associated with chlorine, the experiment was carried out in a well ventilated area. It should be noted that as chlorine is produced, the pH of the cell rises, and the resulting hydroxide ions are able to sequester chlorine by reacting with it to form hypochlorite (The Chlorine Institute, n.d.; Chernev, 2000). Both the the hypochlorites and hydroxides can be harmful however, so safety goggles and gloves were worn to prevent chemical burns, even though the solutions were relatively dilute. Eye protection was also worn while testing the potassium chlorates oxidizing capabilities, and this was done well away from any flammable materials. The electricity used in this experiment was of low voltage so there were no risks associated with it.

To produce minimal environmental impact all chemicals were disposed of in an appropriate manor. There are no serious ethical considerations associated with the conduction of this experiment.

3.3 Preparing the Electrolyte

To prepare the electrolyte, an excess of both potassium chloride and potassium chlorate was dissolved in one liter of water with heat and stirring in a flat-bottom boiling flask. The solution was then allowed to cool to 20°C and filtered to separate any remaining crystals.

The electrolyte was saturated with both potassium chloride and chlorate so that as more potassium chlorate formed, it would crystallize out right away. This allowed for more accurate yield calculations. The electrolyte for all trials was prepared in one batch, ensuring that ion concentrations would be controlled throughout trials.

Using solubility product constants, the concentration of each ion in the electrolyte can be calculated. These concentrations are 4.604M for potassium, 4.509M for chloride, and 0.09534M for chlorate. The calculation is shown in the Appendix.

3.4 Selecting the Anode

As the anode of a chlorate cell is subject to extreme oxidizing conditions and high pH, choices for anode material are rather limited. The main four are graphite, mixed metal oxide (usually iridium and ruthenium oxides), lead dioxide, and platinum (Chernev, 2000). Both mixed metal oxide and lead dioxide anodes are very hard to purchase, and while lead dioxide

anodes can be made, I felt it was not safe to do so (ibid). Initially, a cell was made using a graphite anode, but it quickly corroded, turning the electrolyte black. Because of this, a platinum coated titanium mesh anode was selected for the cell because it was relatively easy to purchase and would offer good corrosion resistance with high currents (ibid). The titanium substrate also allows for high surface areas with low amounts of platinum, and it forms a passive oxide coating, preventing corrosion if the platinum coating fails (Dubent, n.d.).

3.5 Construction of the Cell

The cell was constructed using a 100ml graduated cylinder to hold the electrolyte. This relatively small volume allowed fast trials to be conducted, thus enabling more data collection. The platinum coated titanium anode and a copper cathode were held inside the tube with the use of clamps. A thermometer was placed in the top of the cell to measure cell temperature. The cell is pictured below:

Figure 1: The Chlorate Cell



3.6 Trials

For each trial, 100ml of the previously prepared electrolyte was added to the cell and current was turned on. Current was changed in 0.50A increments between trials, ranging from 0.50A to 3.50A. The cell was then run for the calculated amount of time required for 200000C of charge to pass through it.

A seed crystal of potassium chlorate was added to the bottom of the cell so that potassium chlorate would crystallize on it instead of the anode. The mass of the seed crystal was later

subtracted from the yield.

After the cell was operated for the calculated time, it was cooled to room temperature. This was done so that all of the chlorate formed during the trial would precipitate. The crystals were then filtered, washed once with ice cold water so as to remove chloride contamination, and air dried at room temperature for two days before weighing.

It should be noted that the trials were carried out in random order instead of, for example, from highest current to lowest current. This ensured that any wear on the electrodes didn't contribute to systematic error present in the data.

4 Qualitative Observations and Data

When current in the cell was turned on, gas was produced at both electrodes, with more gas being produced at the cathode. Initially the smell of chlorine was observed, but it subsided as the cell ran, likely due to the mechanism previously described in Section 3.2. After some time the electrolyte took on a slight green colour which was attributed to the presence of hypochlorite. White crystals of potassium chlorate could be seen precipitating at the base of the graduated cylinder, growing in size over the course of the cell's operation. For all trials, the final yield of potassium chlorate was coarse, crystalline, and clear, suggesting high purity. It was confirmed to be potassium chlorate due to its strong oxidizing tendencies when mixed with sugar and lit with a splint. The potassium chlorate produced from one trial is pictured below.

Figure 2: The Yield of Potassium Chlorate



The following is a chart including all the raw data gathered through experimentation:

Table 1: Data

Current (A $\pm$ 0.01)	Voltage (V)	Temperature ($^{\circ}$ C $\pm$ 0.5)	Yield (g $\pm$ 0.001)
0.50	2.83 $\pm$ 0.05	25.0	16.768
1.00	2.99 $\pm$ 0.04	29.0	15.978
1.50	3.25 $\pm$ 0.09	36.0	17.956
2.00	3.33 $\pm$ 0.15	43.0	20.421
2.50	3.53 $\pm$ 0.11	47.0	20.899
3.00	3.71 $\pm$ 0.12	52.0	20.009
3.50	3.93 $\pm$ 0.11	56.0	19.184

5 Data Processing

5.1 Current Efficiency

Current efficiency was calculated by dividing the number of electrons used to complete the desired reaction by the number of electrons that entered the cell. This process may be summarized by the following equation, where E_c is current efficiency, m is the mass of potassium chlorate formed, n is the number of electrons needed to make a molecule of potassium chlorate, F is the Faraday constant, Q is the amount of charge entering the cell, and m_{molar} is the molar mass of potassium chlorate:

$$E_c = \frac{mnF}{Qm_{molar}}$$

The mass of potassium chlorate yielded from each trial was substituted into the above equation to give current efficiency. For example, below is the calculation carried out for the 0.50A trial.

$$E_c = \frac{16.768 \times 6 \times 96485}{200000 \times 122.55} = 0.39605 \pm 0.00002 = 39.605 \pm 0.002\%$$

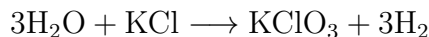
Propagation of uncertainty was also taken into account while calculating current efficiency. This was done using standard procedures, adding either the relative or absolute uncertainties when necessary.

5.2 Energy Efficiency

The data was also processed to give energy efficiency. This was done by finding the Gibbs free energy change associated with the production of the yielded chlorate, and dividing that

by the total energy entering the cell.

First, the Gibbs free energy change needs to be calculated. Assuming standard conditions, a reasonable approximation in this case, standard Gibbs free energies of formation may be used. The standard Gibbs free energy change of reaction ΔG_{rxn}^o is equal to the Gibbs free energy of formation for the products, minus that of the reactants. As mentioned before, the formation of potassium chlorate in an electrolysis cell ideally follows the below equation:



The Free energies of formation for potassium chloride, potassium chlorate, and water are -408.5kJmol^{-1} , -296.3kJmol^{-1} , and -237.1kJmol^{-1} respectively (Engineering ToolBox, 2017). Given this information, the Gibbs free energy change associated with this reaction can be calculated:

$$\Delta G_{rxn}^o = -296.3 - (-408.5 + 3 \times -237.1) = 823.5\text{kJmol}^{-1}$$

This value gives the Gibbs free energy change associated with forming one mole of potassium chlorate. Therefore, multiplying this value by the number of moles potassium chlorate formed will give the total Gibbs free energy change. This total Gibbs free energy change can then be divided by the total amount of energy that entered the cell, which is equal to the voltage times charge. These operations are summarized by the equation below:

$$E_e = \frac{m\Delta G_{rxn}^o}{QV m_{molar}}$$

Data from each trial was substituted into this equation. For example, below is the calculation for the 0.50A trial.

$$E_e = \frac{16.768 \times 823610}{200000 \times 2.83 \times 122.55} = 0.19910 \approx 19.9 \pm 0.4\%$$

Once again, propagation of error was considered using standard procedures.

5.3 Processed Data

Here the processed data is presented. The error bars in Figure 3 are too small to be observed.

Table 2: Data			
Current (A $\pm$ 0.01)	Voltage (V)	Current Efficiency (%)	Energy Efficiency (%)
0.50	2.83 ± 0.05	39.605 ± 0.002	19.9 ± 0.4
1.00	2.99 ± 0.04	37.739 ± 0.002	17.9 ± 0.3
1.50	3.25 ± 0.09	42.411 ± 0.002	18.3 ± 0.5
2.00	3.33 ± 0.15	48.233 ± 0.002	20.2 ± 0.9
2.50	3.53 ± 0.11	49.362 ± 0.002	20.0 ± 0.6
3.00	3.71 ± 0.12	47.260 ± 0.002	18.2 ± 0.6
3.50	3.93 ± 0.11	45.311 ± 0.002	16.5 ± 0.5

Figure 3: Current Efficiency vs. Current

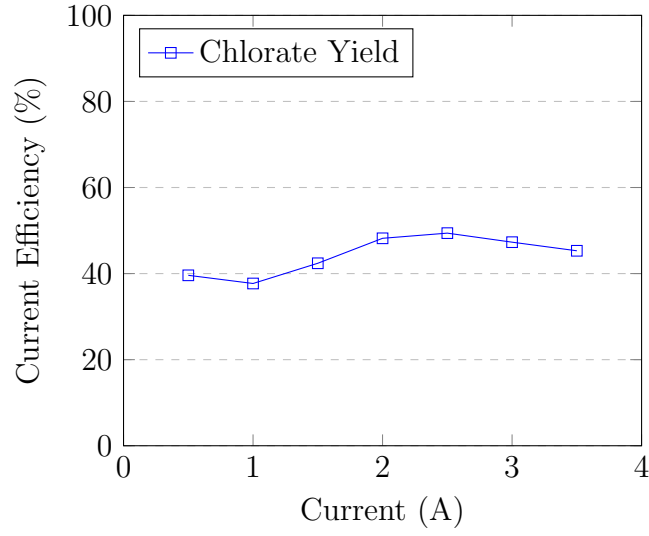
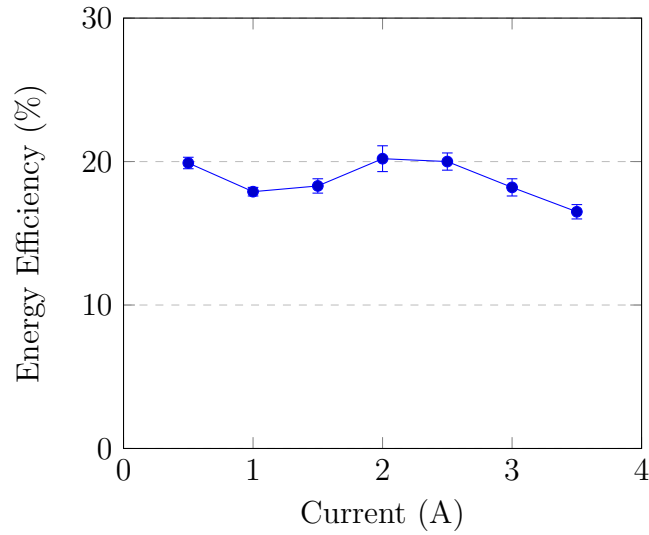


Figure 4: Energy Efficiency vs. Current



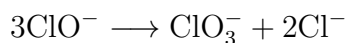
6 Conclusion

As can be observed, the graphs in both Figure 3 and Figure 4 have optimal efficiencies in the middle of their range. On either side of the optimal current, efficiency tends to drop off aside from an increase in the 0.5A trial. Answering the research question "What is the optimal current to produce potassium chlorate with maximum efficiency in an electrolysis cell?", optimal current efficiency of $49.362 \pm 0.002\%$ occurred at 2.50A, and optimal energy efficiency of $20.2 \pm 0.9\%$ occurred at 2.00A. It is important to remember that because discrete current intervals of 0.5 amperes were used in this experiment, the optimal efficiencies likely aren't exactly the values previously given and they should contain an uncertainty of $\pm 0.5A$.

While these were the optimal currents for the cell used in this investigation, the optimal current for other cells may vary. It is suspected that optimal efficiency is really an effect of voltage, rather than current, as electrical potential is generally what determines which reactions can take place (ChemEurope, n.d.). The optimal voltages determined through this experiment are $3.5 \pm 0.2v$ for current efficiency, and $3.3 \pm 0.2v$ for energy efficiency. Once again because of the discrete current intervals in this experiment these values contain large uncertainty.

Initially it was expected that current efficiency would drop with an increase in current due to the extra electrical potential available to carry out side reactions. However this is not the trend that was observed. Instead, current efficiency tended to increase with an increase in current. It is believed that this is because the cells operating at higher currents had higher temperatures.

One of the ways that potassium chlorate is generated in an electrolysis cell is through the disproportionation of hypochlorite:



The above reaction proceeds much faster at higher temperatures (Health Canada, 2008). In fact, potassium chlorate can be produced from solutions of hypochlorite such as bleach and I've done so before. Because of this, the cells operated at higher temperatures were able to convert more hypochlorite into chlorate and thus were higher yielding. These cells that operated at higher temperatures were the ones with higher currents. This is because at higher currents, more power is entering the cell. Since a fraction of this power goes towards heating the cell, more power results in more heat production. Industrial chlorate cells are

heated to take advantage of this effect and improve efficiency (Chernev, 2000).

Another possible reason that the current efficiency went up with current may be that at higher currents/voltages, there is more electrical potential available to perform more energy intensive reactions. In a chlorate cell there are two major reactions: the formation of chlorate from chloride, and the decomposition of water into hydrogen and oxygen gas. The E^0 value for the decomposition of water is -1.23v (Chemistry LibreTexts,n.d.). The E^0 value for the formation of chlorate can be calculated using the below equation (Jay, n.d.):

$$E^0 = -\frac{\Delta G_{rxn}^o}{nF}$$

Plugging in known values:

$$E^0 = -\frac{823610}{6 \times 96485} = -1.42\text{v}$$

The formation of chlorate requires a higher E^0 value than the decomposition of water. Therefore, it is possible that at higher voltages, since the electrons have more energy, a higher fraction of them complete the reaction which requires more energy - or the formation of chlorate. This would have the effect of increasing current efficiency at higher currents.

6.1 Possible Sources of Error

All measurement devices in this investigation used to calculate current efficiency had very high precision, meaning that uncertainty in these calculations was practically zero. The voltage of the cell fluctuated throughout the trials however, meaning that the energy calculations, which depended on voltage readings, were based off of average values with large uncertainties.

To mitigate this source of uncertainty, a possible improvement to the methodology could be to make the cell voltage controlled while monitoring the current continuously. This current monitoring could be done using an electrical ammeter that records the values it measures automatically. Total charge passing through the cell could then be calculated using software, as it is the area under a current vs. time graph, allowing charge to still be controlled between trials. Ideally, in such a setup, the voltage control for the power supply would be done using a voltmeter as close to the electrodes as possible, instead of the one inside the supply itself. This would compensate for the voltage drop between the power supply and the cell.

A possible source of random error in this experiment could be mechanical loss during the

filtering and weighing of the product. While great effort was made to reduce this source of error, unfortunately it is practically unavoidable.

A possible source of systematic error could have come from the washing of the potassium chlorate before drying. Because the rate of chlorate production was higher in the higher current cells, the potassium chlorate would precipitate faster and thus would have a smaller crystal size (MIT, n.d.). This means that it would have a higher surface area. When the washings were performed, the chlorate with a higher surface area, which was the chlorate made in the higher current cells, would dissolve faster. This could have the effect of lowering the yields for the higher current cells, thus introducing systematic error.

While it was decided not to control temperature for this investigation, for future experiments temperature control may have benefits. This could be done by wrapping the existing apparatus with nichrome wire and insulation. Temperature could then be monitored with an electrical thermocouple and when it drops below a certain temperature, current could be run through the nichrome wire to heat the cell. This way the cell could be maintained at a temperature of, for example, 50°C for all trials. This would allow for the effects of increased current/voltage in the cell to be differentiated from those of increased temperature.

6.2 Summary

This experiment proved to be an effective means to gather data to answer the question “What is the optimal current to produce potassium chlorate with maximum efficiency in an electrolysis cell?”. In the cell used for this investigation, optimal current efficiency occurred at $2.5 \pm 0.5\text{A}$ and $3.1 \pm 0.2\text{v}$, whereas optimal energy efficiency occurred at $2.0 \pm 0.5\text{A}$ and $3.3 \pm 0.2\text{v}$. While it was difficult to discern whether these optimal efficiencies were caused by current or temperature, it was still possible to draw conclusive results on the optimal current for the electrolysis cell used in this experiment. The findings of this investigation have applications in the areas of science and engineering, such as in the production of potassium chlorate, where they enable better energy efficiencies and therefore lower costs and environmental impact. Through the conduction of this investigation, relevant scientific information gathered, thus enhancing understanding in the fields of electrochemistry and in science as a whole.

7 Works Cited

- American Chemical Society. (2019, December 23). *Potassium chlorate*. ACS Chemistry For Life. Retrieved March 3, 2024, from <https://www.acs.org/molecule-of-the-week/archive/p/potassium-chlorate.html>
- ChemEurope. (n.d.). Electrolysis. ChemEurope. Retrieved March 4, 2024, from <https://www.chemeuropa.com/en/encyclopedia/Electrolysis.html>
- Chemistry LibreTexts. (n.d.). Calculating Standard Cell Potentials. Chemistry LibreTexts. Retrieved March 4, 2024, from [https://chem.libretexts.org/Bookshelves/Introductory_Chemistry/Introductory_Chemistry_\(CK-12\)/23%3A_Electrochemistry/23.06%3A_Calculating_Standard_Cell_Potentials](https://chem.libretexts.org/Bookshelves/Introductory_Chemistry/Introductory_Chemistry_(CK-12)/23%3A_Electrochemistry/23.06%3A_Calculating_Standard_Cell_Potentials)
- Chernev, D. (2000, January 1). Anodes. ExRockets. Retrieved March 3, 2024, from <http://chlorates.exrockets.com/anodes.html>
- Dubent, S., & Mazard, A. (n.d.). *Characterization and corrosion behaviour of grade 2 titanium used in electrolyzers for hydrogen production*. Sciencedirect. <https://www.sciencedirect.com/science/article/pii/S0360319919315228>
- Engineering ToolBox. (2017). *Standard State and Enthalpy of Formation, Gibbs Free Energy of Formation, Entropy and Heat Capacity*. Engineering ToolBox. Retrieved March 3, 2024, from https://www.engineeringtoolbox.com/standard-state-enthalpy-formation-definition-value-Gibbs-free-energy-entropy-molar-heat-capacity-d_1978.html
- The Engineering ToolBox (2017). *Inorganic Compounds in Water - Melting and Boiling Temperature, Density and Solubility*. Engineering ToolBox. Retrieved March 3, 2024, from https://www.engineeringtoolbox.com/inorganic-salt-melting-boiling-point-water-solubility-density-liquid-d_1984.html
- EPN. (2019). *Sodium Chlorate Supply Chain – Executive Summary*. EPN. <https://www.epa.gov/system/files/documents/2023-03/Sodium%20Chlorate%20Supply%20Chain%20Profile.pdf>
- Harvard. (2015). *Lecture 1: Pre-reading*. Harvard. <file:///home/tom/Downloads/01-27-L1->

PreReading.pdf

Health Canada. (2008). *Chlorite and Chlorate*. Health Canada. <https://www.canada.ca/content/dam/canada/health-canada/migration/healthy-canadians/publications/healthy-living-vie-saine/water-chlorite-chlorate-eau/alt/water-chlorite-chlorate-eau-eng.pdf>

Jay. (n.d.). *Standard potential, free energy, and the equilibrium constant*. Khan Academy. Retrieved March 4, 2024, from <https://www.khanacademy.org/science/ap-chemistry-beta/x2eef969c74e0d802:applications-of-thermodynamics/x2eef969c74e0d802:cell-potential-and-free-energy/v/standard-potential-free-energy-and-the-equilibrium-constant>

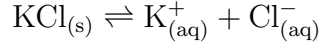
MIT Department of Chemistry. (n.d.). *Growing Quality Crystals*. X-Ray Diffraction Facility. Retrieved March 4, 2024, from <https://web.mit.edu/x-ray/crystallize.html>

NurdRage. (2019, January 15). *Make Potassium Chlorate by Electrolysis - The Basic Guide* [Video]. YouTube. <https://www.youtube.com/watch?v=itAnVJ2vpHQ>

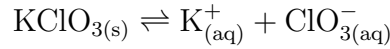
Stanford. (n.d.). *What Makes Things Electrical?* Stanford. <https://web.stanford.edu/class/archive/engr/engr40m.1178/reader/chapter1.pdf>

8 Appendix

The concentration of each of the ions in the electrolyte can be calculated using the solubility product constant, K_{sp} . First, the solubility product constants can be calculated for both potassium chloride, and chlorate:



$$K_{spKCl} = [\text{K}^{+}][\text{Cl}^{-}]$$



$$K_{spKClO_3} = [\text{K}^{+}][\text{ClO}_3^{-}]$$

At 20°C the solubilities of potassium chloride and chlorate are 339.7gL⁻¹ and 81.5gL⁻¹ respectively (Engineering ToolBox, 2017). Dividing by their molar masses, we get their molar solubilities: 4.556M and 0.6626M respectively. Knowing that these are the concentrations of each of their constituent ions, the solubility product constant can be calculated.

$$K_{spKCl} = 4.556^2 = 20.76$$

$$K_{spKClO_3} = 0.6626^2 = 0.4390$$

We also know that the concentration of potassium ions is equal to the concentration of chloride ions, plus that of chlorate ions due to the fact that each time a chloride or chlorate ion dissolves, so does a potassium ion. Therefore:

$$[\text{K}^{+}] = [\text{Cl}^{-}] + [\text{ClO}_3^{-}]$$

$$[\text{Cl}^{-}] = [\text{K}^{+}] - [\text{ClO}_3^{-}]$$

$$K_{spKCl} = [\text{K}^{+}](\text{K}^{+} - [\text{ClO}_3^{-}])$$

$$K_{spKCl} = [\text{K}^{+}]^2 - [\text{K}^{+}][\text{ClO}_3^{-}]$$

$$K_{spKCl} = [\text{K}^{+}]^2 - K_{spKClO_3}$$

$$[\text{K}^{+}] = \sqrt{K_{spKCl} + K_{spKClO_3}} = 4.604\text{M}$$

$$[\text{Cl}^{-}] = \frac{K_{spKCl}}{[\text{K}^{+}]} = 4.509\text{M}$$

$$[\text{ClO}_3^{-}] = \frac{K_{spKClO_3}}{[\text{K}^{+}]} = 0.09534\text{M}$$